

**Online therapy and face-to-face therapy preferences for individuals based on
extroversion/introversion personality traits**

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Abstract

As the prevalence of depression and anxiety continues to increase, it is important to expand our knowledge on variables that can impact accessibility and preferences of various therapy forms. There remains a major gap in literature in regards to how personality characteristics can impact preferences for forms of therapy. The primary aim of this study was to assess how introversion and extroversion can impact preferences between online therapy and face-to-face therapy. I hypothesized that (1) participants who were identified as introverted would have a stronger preference for online therapy, and (2) participants who were identified as extroverted would have a stronger preference for face-to-face therapy. Participants (N = 45; Mean age = 20.08) were assessed based on their discomfort and value scores obtained for both online and face-to-face therapy. Introverts reported higher discomfort for both online and face-to-face therapy than extroverts. Participants reported lower value scores for online therapy than face-to-face therapy.

Keywords: introversion, extroversion, online therapy, face-to-face therapy, preferences

**Online therapy vs. face-to-face therapy preferences for individuals based on
extroversion/introversion personality traits**

Although we have seen a great deal of improvement in psychotherapeutic and psychopharmacological treatments for mental health, the prevalence of depression around the world continues to increase. It is estimated that by 2030, depressive disorders will be the second leading cause of death worldwide (Mathers & Loncar, 2006). Depression impacts such a large portion of the population, yet many individuals go untreated and undiagnosed (Shim et al., 2011). During 2020 it was reported that prevalence of anxiety had increased by three times and prevalence of depression had increased by four times in comparison to 2019 (Czeisler et al., 2020). The prevalence of depressive episodes has increased among the adolescent population, but there has been no progress observed in narrowing the treatment gap of those experiencing depression (Moitabai et al., 2016). Although it is important to research how depression impacts all categories of individuals, university students are one of the most understudied groups, yet also one of the most steadily increasing populations impacted by depression. In contrast to the rate of depression observed in the general population, university students have an average depression prevalence of 30.6% (Ibrahim et al., 2013).

In our evolving digital world, a potential way to increase access to mental health treatment is through online therapy. Benefits of online therapy include allowing the individual to feel more secure and open to disclosing information, less of a stigma associated with it, reduces waiting time for therapy, allows a wider range of therapists to choose from, and is more available and affordable than traditional face-to-face therapy (Amichai-Hamburger et al., 2014). Previous research has shown that internet based cognitive behavioral therapy (ICBT) has been effective in reducing symptoms of depression, anxiety, and insomnia (Topooco et al., 2018; Ye et al., 2015).

According to a survey conducted by the American Psychiatric Association, prior to the pandemic 63.6% of therapists reported not using any online therapy sessions in comparison to the 1.9% of therapists who did not use online therapy sessions after the onset of the pandemic. It was also reported that 81% of therapists have continued to see 75-100% of patients through online therapy (American Psychiatric Association, 2020). While this change was not originally by choice, it is clear that many patients are accepting of this new form of therapy and it is important to understand how this could be used to expand access to mental health treatment.

A recent research study examined the effectiveness of internet delivered therapy for anxiety and depression in university students. The participants in this study participated in a 5-week cognitive behavioral therapy online program. The results of the study showed clinically significant reduction in reported symptoms of both anxiety and depression. The study also found high acceptability of the online therapy, with seventy-five percent of participants reporting being satisfied (Dear et al., 2019).

Another study evaluated the effectiveness of ICBT compared to a control condition in treatment of adolescent depression. This study randomly assigned participants to either a control group or an 8-week internet delivered therapy session program. The results of the study showed significant reduction in depressive symptoms in the ICBT group and the improvement was maintained at six months post treatment (Topooco et al., 2018).

The use of online therapy is relatively new so there is little research on whether it is more effective in treating depression in college students than traditional face-to-face therapy. With the increase in technology use for both personal, professional, and health care reasons in this generation, it would be beneficial to examine what people's preferences are for using technology as a mode of therapy or for traditional modes of therapy. Previous studies have looked into

general preferences and perceived benefits of online therapy in comparison to face-to-face therapy (Wong et al., 2018; Apolinario-Hagen et al., 2018). Yet, there remains a major gap in literature in regards to how personality characteristics can impact the preferences and perceived effectiveness of online therapy on an individual basis.

This study could potentially give us more knowledge on differences in preference for therapy between introverts and extroverts. This research could also be beneficial towards college campuses in adding an online therapy option for those who would prefer that, in addition to traditional face-to-face therapy, which could improve accessibility to therapy for more individuals. Introverts are characterized by a focus on internal feelings rather than on external sources of stimulation. Often introverts are “withdrawn, reserved, quiet, and deliberate; they may guard expression of positive affect and prefer to work independently” (American Psychological Association, n.d.). As a result of these characteristics it is expected that participants who are identified as introverted will have a stronger preference for online therapy (Hypothesis 1). Extroverts are characterized by a focus on obtaining gratification from what is outside the self and engaging socially with others. Often extroverts are “outgoing, sociable, openly expressive” and prefer to work with others (American Psychological Association, n.d.). As a result of these characteristics it is expected that participants who are identified as extroverted will have a stronger preference for face-to-face therapy (Hypothesis 2).

Method

Participants

A total of 46 college students from The College of St. Scholastica participated in this study. The demographics of the sample of participants are as follows: N = 46 (43 (93.5%) women; age range 18-31, mean = 20.08 (SD = 2.373); 40 (86.96) white; 20 (43.5%) introverted.

See Table 1 and Figure 3 for demographic characteristics. The participants were selected based on a convenience sample of students enrolled at The College of St. Scholastica, and their willingness/availability to participate in the study. Participants were recruited via the college community email platform to reach potential students available to participate in the study. Students under the age of 18 were excluded from the study. No vulnerable populations were targeted.

Materials

The demographics questionnaire consisted of six basic demographic questions. Examples of these questions include: age, level of education, race, ethnicity, sex at birth, and gender.

The extroversion/introversion scale taken from the Big Five Scale (Goldberg, 1992) is a 20 item self-report questionnaire that measures levels of extroversion and introversion on a 5-point scale from 1 (strongly disagree) to 5 (strongly agree). Example items from the Big Five would be: “I don’t mind being the center of attention” and “I often feel uncomfortable around others”.

The Face-to-Face Counseling Attitude Scale (FCAS; Rochlen et al., 2004) measures attitudes towards face-to-face therapy on a 5-point scale from 1 (strongly disagree) to 5 (strongly agree). Attitudes towards the therapy are scored based on five questions which measure discomfort level, and five questions which measure levels of value. Example items from the FCAS would be: “If I were having a personal problem, seeking help with a face-to-face counselor would be the last option I would consider” and “Using face-to-face counseling would help me learn about myself”.

The Online Counseling Attitudes Scale (OCAS; Rochlen et al., 2004) measures attitudes towards online therapy on a 5-point scale from 1 (strongly disagree) to 5 (strongly agree).

Attitudes towards the therapy are scored based on five questions which measure discomfort level, and five questions which measure levels of value. Example items from the OCAS would be: “If I were having a personal problem, seeking help with an online counselor would be the last option I would consider” and “Using online counseling would help me learn about myself”.

Procedure

After participants were recruited through the college community email platform, they completed an anonymous online survey (Qualtrics). Names, email addresses, and IP addresses were not collected. Participants remained anonymous throughout the entire study. Once participants provided consent, they then provided basic demographic information. The demographics questionnaire asked participants their age, gender, race/ethnicity, and academic status (i.e. first year, fourth year). Participants then completed the remaining portions of the study. This included answering questions specific to the study (preferences for online and face-to-face therapy), and completing a personality scale. Questions specific to this study only collected attitudes towards online and face-to-face therapy, not the participants’ personal experiences with therapy. This study took participants no more than 15 minutes to complete. Upon completion of the study, participants were thanked for their participation.

Ethics Board Approval Statement

This project was reviewed and approved by the Institutional Review Board of The College of St. Scholastica.

Results

Out of the initial 46 participants who completed the study, 11 participants (23.91%) scored in the neutral range on the Big Five Scale. To conduct a comparative analysis between

introverted and extroverted personality types, these neutral participants were excluded from the 2 x 2 ANOVA models, resulting in a final sample size of $N = 35$ (20 introverts, 15 extroverts).

Online Discomfort and Face-to-Face Discomfort

A 2 x 2 mixed-design ANOVA was calculated to examine the effects of therapy type (online and face-to-face) and personality type (introvert and extrovert) on therapy discomfort scores (see Figure 1 and Table 2). The therapy type x personality type interaction ($F(1,33) = 0.77$, $p = 0.387$) and the main effect for therapy type ($F(1,33) = 3.727$, $p = 0.062$) were not significant. In addition, a main effect for personality type was significant ($F(1,33) = 5.734$, $p = 0.022$). Introverts ($M = 14.10$, $SD = 5.098$) reported higher discomfort for online therapy than extroverts ($M = 11.07$, $SD = 2.915$). Introverts ($M = 11.80$, $SD = 4.927$) also reported higher discomfort for face-to-face therapy than extroverts ($M = 10.00$, $SD = 2.563$).

Online Value and Face-to-Face Value

A 2 x 2 mixed-design ANOVA was calculated to examine the effects of therapy type (online and face-to-face) and personality type (introvert and extrovert) on therapy value scores (see Figure 2 and Table 3). The therapy type x personality type interaction ($F(1,33) = 2.273$, $p = 0.141$) and the main effect of personality type ($F(1,33) = 1.128$, $p = 0.296$) were not significant. In addition, a main effect of therapy type was significant ($F(1,33) = 12.498$, $p = 0.001$). Participants reported lower value scores for online therapy ($M = 18.83$, $SD = 3.899$) than face-to-face therapy ($M = 21.17$, $SD = 2.595$).

Discussion

This study aimed to further the understanding of potential relationships between personality characteristics, specifically extroversion and introversion, and attitudes towards different therapy types. Results found that there was a significant relationship between

personality type and discomfort scores, indicating that introverts reported higher discomfort for both online and face-to-face therapy than extroverts. While therapy type did not have a significant relationship with discomfort scores, it does indicate a trend in which participants, regardless of personality type, reported greater discomfort for online therapy than face-to-face therapy. Results also found a significant relationship between therapy type and value scores, indicating that participants reported lower value scores for online therapy than face-to-face therapy. The overall goal of this study was to look into preferences for therapy type, and in this study ‘preferences’ was operationalized in terms of discomfort and value. Stronger preferences would be indicated by higher value scores and lower discomfort scores. The hypothesis that introverts would have a stronger preference for online therapy was not supported considering results found lower mean scores for value, and higher mean scores for discomfort. This unexpected finding may be attributed to unique virtual dynamics such as unnatural eye contact and hyper awareness of one’s self-image on camera. For introverts who are already prone to social monitoring, virtual interaction might generate greater discomfort than sitting in a physical room with a therapist. The hypothesis that extroverts would have a stronger preference for face-to-face therapy was partially supported considering results found higher mean scores for value, and lower mean scores for discomfort.

To the best of my knowledge, this is the first study to look into the relationship between personality and therapy type on the specific attitudes of discomfort and value. Therefore, the resulting relationships, or lack of relationships, found in this study present relevant preliminary information towards this new topic.

One of the biggest strengths of this study is that it is relevant and, to the best of my knowledge, the first to examine the relationships between personality, therapy type, and attitudes.

The extreme increase in use of online therapy during and post COVID pandemic highlights the importance of this study to understand what factors can relate to attitudes surrounding the various forms of therapy. A major limitation is the lack of demographic variability and small sample size. Considering only 46 participants, all of which are college students at The College of St. Scholastica, and almost all of which were white and women, this study cannot be generalized to the general population. Additionally, while there were 46 participants, categorizing personality types required dropping 11 'neutral' participants to examine introvert and extrovert groups specifically. This further reduced the statistical power and highlights the need for larger initial sample sizes in future research. Another weakness of this study is the potential confound of the COVID pandemic. This event could have had a significant relationship with participants' use of online therapy, and/or a general increase in use of therapy. Another limitation is that this study was non-experimental, so we can only look at the results in terms of relationships rather than as a causal attribution. Another limitation to this study is that preferences were scored based on attitudes of discomfort and value, so any potential relationships that could exist in relation to other types of attitudes are unknown at this time.

Future studies could replicate this present study in order to examine the results from a larger, more diverse sample of participants to further the knowledge found in these current results. Future research could also benefit from looking into the relationship that various cultures and sociodemographic groups could have on attitudes towards therapy. Another possible area for future research would be to score 'preferences' using things in addition to and/or different from the attitudes of discomfort and value like this study chose to do. Considering the limitation of this study being unable to draw conclusions of causality, it would also be beneficial for future research to consider using an experimental design.

Conclusion

This study aimed to provide preliminary knowledge of the relationship between personality, therapy type, and preferences. Contrary to expectations, introverts did not show a preference for online therapy. Results found that introverts reported higher discomfort across both online and face-to-face therapy settings compared to extroverts. Furthermore, participants overall perceived face-to-face therapy as higher in value than online therapy. While limited by sample size and demographic homogeneity, these findings remain relevant and provide new knowledge for future research into therapy preferences. Therapy is widely used, and it is important for mental health providers to consider personality dynamics and individual comfort levels when recommending therapy modalities.

References

- American Psychiatric Association. (2020). *Psychiatrists use of telepsychiatry during COVID-19 public health emergency*. APA.
<https://www.psychiatry.org/File%20Library/Psychiatrists/Practice/Telepsychiatry/APA-Telehealth-Survey-2020.pdf>
- American Psychological Association. (n.d.). Extraversion. In *American Psychological Association dictionary*. Retrieved March 2, 2022, from
<https://dictionary.apa.org/extraversion>
- American Psychological Association. (n.d.). Introversion. In *American Psychological Association dictionary*. Retrieved March 2, 2022, from
<https://dictionary.apa.org/introversion>
- Amichai-Hamburger, Y., Klomek, A., Friedman, D., Zuckerman, O., & Shani-Sherman, T. (2014). The future of online therapy. *Computers in Human Behavior*, 41, 288-294.
<https://doi.org/10.1016/j.chb.2014.09.016>
- Apolinario-Hagen, J., Harrer, M., Kahlke, F., Fritsche, L., Salewski, C., & Ebert, D. D. (2018). Public attitudes toward guided internet-based therapies: Web-based survey study. *JMIR Mental Health*, 5(2). <https://doi.org/10.2196/10735>
- Czeisler, M. E., Lane, R. I., Petrosky, E., Wiley, J. F., Christensen, A., Njai, R., Weaver, M. D., Robbins, R., Facer-Childs, E. R., Barger, L. K., Czeisler, C. A., Howard, M. E., Rajaratnam, S. M. W. (2020). Mental health, substance use, and suicidal ideation during the COVID-19 pandemic— United States, June 24-30, 2020. *Morbidity and Mortality Weekly Report*, 69, 1049-1057. <http://dx.doi.org/10.15585/mmwr.mm6932a1>

- Dear, B., Johnson, B., Singh, A., Wilkes, B., Brkic, T., Gupta, R., Jones, M., Bailey, S., Dudeney, J., Gandy, M., Fogliati, R., & Titov, N. (2019). Examining an internet-delivered intervention for anxiety and depression when delivered as a part of routine care for university students: A phase IV trial. *Journal of Affective Disorders*, 256, 567-577. <https://doi.org/10.1016/j.jad.2019.06.044>
- Goldberg, L. R. (1992). The development of markers for the Big-Five factor structure. *Psychological Assessment*, 4(1), 26–42. <https://doi-org.akin.css.edu/10.1037/1040-3590.4.1.26>
- Ibrahim, A. K., Kelly, S. J., Adams, C. E., & Glazebrook, C. (2013). A systematic review of studies of depression prevalence in university students. *Journal of Psychiatric Research*, 47(3), 391-400. <https://doi.org/10.1016/j.jpsychires.2012.11.015>
- Mathers, C., & Loncar, D. (2006). Projections of global mortality and burden of disease from 2002 to 2030. *PLoS Medicine*, 3(11). <https://doi.org/10.1371/journal.pmed.0030442>
- Moitabai, R., Olfson, M., & Han, B. (2016). National trends in the prevalence and treatment of depression in adolescents and young adults. *Pediatrics*, 138(6). <https://doi.org/10.1542/peds.2016-1878>
- Rochlen, A. B., Natasha Beretvas, S., & Zack, J. S. (2004). The Online and Face-to-Face Counseling Attitudes Scales: A Validation Study. *Measurement & Evaluation in Counseling & Development*, 37(2), 95–111. <https://doi-org.akin.css.edu/10.1080/07481756.2004.11909753>
- Shim, R. S., Baltrus, P., Ye, J., & Rust, G. (2011). Prevalence, treatment, and control of depressive symptoms in the United States: Results from the national health and nutrition

- examination survey (NHANES), 2005-2008. *Journal of the American Board of Family Medicine*, 24(1), 33-38. doi:10.3122/jabfm.2011.01.100121
- Topooco, N., Berg, M., Johansson, S., Liljethorn, L., Radyogin, E., Vlaescu, G., Nordgren, L. B., Zetterqvist, M., & Andersson, G. (2018). Chat and internet-based cognitive behavioral therapy in treatment of adolescent depression: Randomized controlled trial. *BJPsych Open*, 4(4), 199-207. <http://dx.doi.org.akin.css.edu/10.1192/bjo.2018.18>
- Wong, K. P., Bonn, G., Tam, C. L., & Wong, C. P. (2018). Preferences for online and/or face-to-face counseling among university students in Malaysia. *Frontiers in Psychology*, 9, 64. <https://doi.org/10.3389/fpsyg.2018.00064>
- Ye, Y., Zhang, Y., Chen, J., Liu, J., Li, X., Liu, Y., Lang, Y., Lin, L., Yang, X., Jiang, X. (2015). Internet-based cognitive behavioral therapy for insomnia (ICBT-i) improves comorbid anxiety and depression— a meta-analysis of randomized controlled trials. *PLoS ONE*, 10(11). <https://doi.org/10.1371/journal.pone.0142258>

Table 1*Demographic Characteristics of Participants*

Demographic Characteristic	<i>n</i>	%
Race & Ethnicity		
American Indian or Alaska Native	0	0
Asian	0	0
Black or African American	0	0
Hispanic or Latino	0	0
Native Hawaiian or Pacific Islander	0	0
Multiracial	6	13.04
White/Caucasion	40	86.96
Prefer to Self-Describe	0	0
Sex at Birth		
Female	43	93.5
Male	3	6.5
Intersex	0	0
Prefer to Self-Describe	0	0
Gender		
Female	43	93.5
Male	3	6.5
Non-Binary	0	0
Transgender	0	0
Prefer to Self-Describe	0	0
Level of Education		
First-Year	12	26.1
Sophomore	9	19.6
Junior	9	19.6
Senior	13	28.3
Graduate Student	2	4.3
Other	1	2.2

Table 2

Descriptive Statistics

	Personality Type	Mean	Std. Deviation	N
Online Discomfort	Introvert	14.10	5.098	20
	Extrovert	11.07	2.915	15
	Total	12.80	4.510	35
Face-to-Face Discomfort	Introvert	11.80	4.927	20
	Extrovert	10.00	2.563	15
	Total	11.03	4.134	35

Table 3

Descriptive Statistics

	Personality Type	Mean	Std. Deviation	N
Online Value	Introvert	18.00	4.565	20
	Extrovert	19.93	2.520	15
	Total	18.83	3.899	35
Face-to-Face Value	Introvert	21.15	2.641	20
	Extrovert	21.10	2.624	15
	Total	21.17	2.595	35

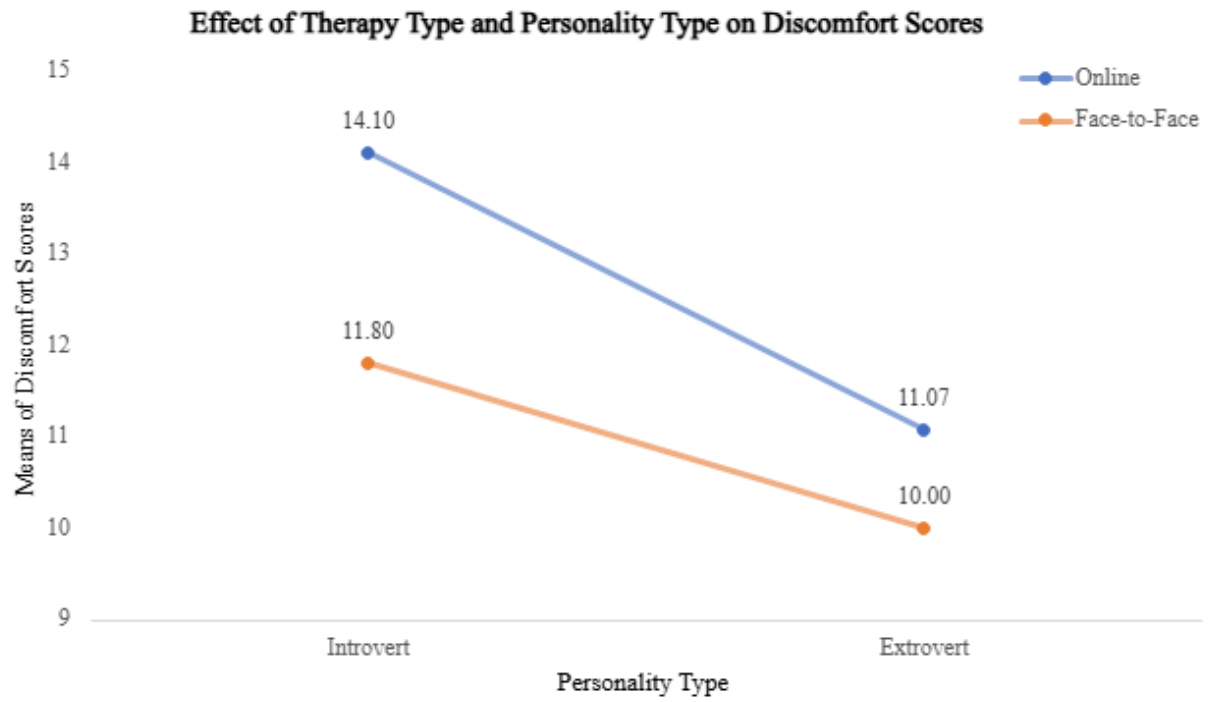
Figure 1

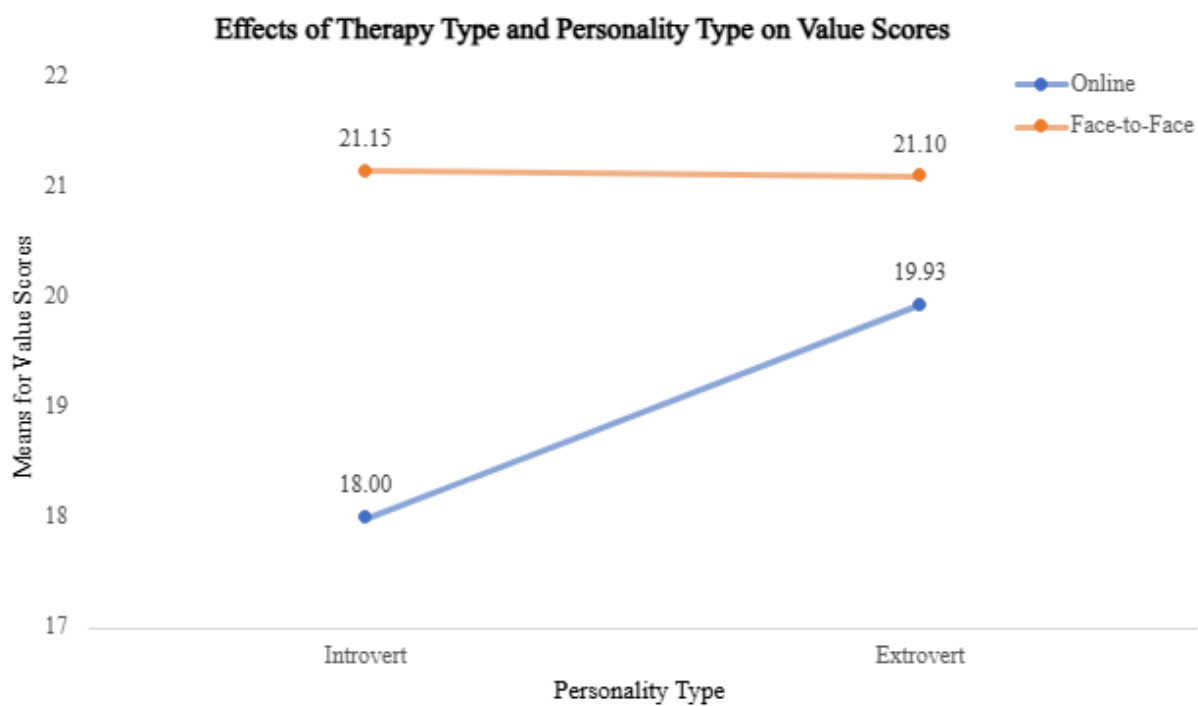
Figure 2

Figure 3